

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ace5d7b1dc238605c.jsonl`  
- SHA-256: `dc680ff4b80afd4956e5b0120ee7e28d03e1b8fea0de6cd9c0e8e9584ba30105`  
- Source modified: `2026-05-31T04:25:31+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:24:27.357Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"The Surya CODE and the Surya WEIGHTS are licensed separately: the repo's code LICENSE file is Apache-2.0 (permissive), but the MODEL_LICENSE explicitly excludes source code/scripts ('Complementary Material') from the OpenRAIL-M terms. Verifying the repo license alone would wrongly conclude Surya is fully permissive ? the weights are the binding constraint, and the current weights license is OpenRAIL-M, NOT the GPL/commercial dual-license that older descriptions cite."

****Source:**** https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)

****Supporting quote:**** "'Complementary Material' means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:24:29.939Z)

I'll verify this claim about Surya's licensing structure. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:24:33.663Z)

I'll fetch the primary source first, then search for contradicting evidence.

4. user (2026-05-31T04:24:39.386Z)

License Analysis

This is an **Apache License 2.0** file.

License Header:

"Apache License Version 2.0, January 2004"

Key Points:

The license permits commercial use. It grants a "perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable copyright license" for reproduction, modification, and distribution.

There is no dual licensing mentioned in this document. Users may add their own license terms for modifications, but the underlying work remains under Apache 2.0.

5. user (2026-05-31T04:24:39.987Z)

MODEL_LICENSE Analysis

License Type: AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1

OpenRAIL-M Mention: Yes, the license explicitly states it is "generally applicable to any machine-learning Model" and references the "Open" and "RAIL" nomenclature.

Complementary Material Definition:

The license defines it as: "the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

Exclusion from License Terms:

The definition concludes: "Complementary Material is not licensed under this License." This means source code and scripts are explicitly excluded from the license coverage.

Commercial-Use Restrictions (Attachment A):

The license restricts use for entities that:

- Generated more than \$5,000,000 in gross revenue in the prior year (except personal/research use)
- Raised more than \$5,000,000 in total equity or debt funding (except personal/research use)
- Provide products/services competing with the licensor's offerings

Use-Based Restrictions:

Per Section 5, users cannot use the Model in violation of restrictions in Attachment A, which prohibit uses that violate laws, infringe IP rights, or breach the commercial restrictions noted above.

6. assistant (2026-05-31T04:24:43.845Z)

The primary source confirms several elements of the claim. However, I notice an important detail: the MODEL_LICENSE is "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" with a **\$5M revenue/funding commercial restriction** ? this is more nuanced than the claim states. The claim says the weights license is "OpenRAIL-M, NOT the GPL/commercial dual-license." Let me verify the historical claim (was it ever GPL?) and confirm the current state.

7. user (2026-05-31T04:24:51.014Z)

Web search results for query: "Surya OCR datalab-to license GPL commercial dual license model weights"

Links: [{"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "surya/MODEL_LICENSE at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya"}]

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to/surya/blob/master/MODEL_LICENSE"},"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"},"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages","url":"https://github.com/datalab-to/surya/tree/master"},"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"},"title":"surya/README.md at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/README.md"},"title":"GitHub - CibirQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub","url":"https://github.com/CibirQuadance/surya-OCR"},"title":"https://github.com/datalab-to/surya | Ecosyste.ms: Awesome","url":"https://awesome.ecosyste.ms/projects/github.com/datalab-to/surya"},"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"},"title":"Datalab develops small AI models to power 'document intelligence' in the enterprise","url":"https://www.forkable.io/p/datalab-develops-small-ai-models"}]
```

Based on the search results, here's what I found about Surya OCR's licensing model:

License Structure

The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$2M funding/revenue) and the code is GPL.

Dual Licensing Model

Surya operates a dual licensing approach:

****Model Weights:**** The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

****Code:**** The code itself is GPL-licensed, fostering community contributions while offering commercial licensing options for enterprises that need to remove GPL requirements.

Commercial Licensing

For broader commercial licensing or to remove GPL requirements, visit their pricing page. Larger companies need a commercial license from Datalab to remove GPL code requirements and access enterprise support.

In summary, Surya implements a dual licensing model where the code is GPL-licensed and the model weights are available under a modified Open Rail-M license for non-commercial use, with commercial licenses available for enterprises that need to remove GPL restrictions or exceed the funding/revenue thresholds.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-05-31T04:24:52.038Z)

Web search results for query: "Surya OCR "OpenRAIL-M" license model weights commercial"

Links: [{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"},"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-ocr/"},"title":"surya/MODEL_LICENSE at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"},"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"},"title":"surya-ocr 0.17.1 on PyPI - Libraries.io - security & maintenance data for open source software","url":"https://libraries.io/pypi/surya-ocr"},"title":"surya/README.md at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/README.md"},"title":"Surya: The Revolutionary OCR Toolkit for 90+

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Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"},"title":"GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub","url":"https://github.com/CibiQuadance/surya-OCR"},"title":"GitHub - cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages · GitHub","url":"https://github.com/cognitiveailab/surya-sci"},"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"}]
```

Based on the search results, here's what I found about Surya OCR's licensing model:

License Summary

The Surya code is licensed under Apache 2.0, while the model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

Commercial Usage

For commercial use beyond the free tier, broader commercial licensing of the model weights is available through their pricing page.

About the OpenRAIL-M License

The OpenRAIL-M license used by Surya is a modified version designed to balance openness with responsible use. The "Open" nomenclature indicates that the licensed Model is freely accessible to downstream and other users, while the "RAIL" nomenclature indicates that there are use restrictions prohibiting the use of the Model.

Key Takeaway

Surya provides free access to model weights for research, personal projects, and startups under \$5M in funding/revenue through the modified OpenRAIL-M license. Organizations exceeding this threshold or those requiring GPL license removal need to pursue commercial licensing options.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:24:55.408Z)

The search results reveal important nuance. The first search says "the code is GPL" while the second says "code is licensed under Apache 2.0." This is the historical transition the claim references. Let me fetch the current README to confirm the present-day licensing statement, since the claim hinges on the *current* state.

10. user (2026-05-31T04:25:00.537Z)

Licensing Information from Surya README

Code License

"The Surya code is licensed under Apache 2.0."

Model Weights License

"The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

Commercial Licensing

The document states: "For broader commercial licensing of the model weights, visit our pricing page here."

****Summary:**** The code uses Apache 2.0, while model weights follow a modified OpenRAIL-M license permitting research and startups with less than \$5M in funding/revenue. Commercial users beyond these parameters need separate licensing.

11. user (2026-05-31T04:25:01.991Z)

License Information for Surya OCR 2

Based on the Hugging Face model card, the license is **OpenRAIL** (specifically a modified AI Pubs Open Rail-M license).

License Metadata

The model card displays:

- **License tag**: `openrail`

Licensing Statement

From the "Commercial usage" section:

> "The Surya code is licensed under Apache 2.0. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue). For broader commercial licensing of the model weights, visit our pricing page [here](https://www.datalab.to/pricing?utm_source=gh-surya)."

Summary

- **Code**: Apache 2.0
- **Model Weights**: Modified AI Pubs Open Rail-M (OpenRAIL-M)
 - Free for: research, personal use, startups under \$5M funding/revenue
 - Commercial licensing available for purchase

12. assistant (2026-05-31T04:25:15.727Z)

I now have strong primary-source confirmation. Let me assess the claim against the evidence.

The claim has multiple sub-assertions. Let me evaluate each:

1. **"Code LICENSE file is Apache-2.0"** ? CONFIRMED. The `LICENSE` file is Apache 2.0; README and HF card both state "The Surya code is licensed under Apache 2.0."
2. **"MODEL_LICENSE excludes source code/scripts ('Complementary Material') from OpenRAIL-M terms"** ? CONFIRMED by the supporting quote. The MODEL_LICENSE explicitly defines Complementary Material and states "Complementary Material is not licensed under this License."
3. **"Verifying repo license alone would wrongly conclude Surya is fully permissive"** ? CONFIRMED. The weights are NOT fully permissive ? they carry use-based restrictions and a \$5M revenue/funding commercial gate.
4. **"The weights are the binding constraint"** ? CONFIRMED. The weights license is the restrictive one; Apache-2.0 code alone is useless without the weights.
5. **"The current weights license is OpenRAIL-M, NOT the GPL/commercial dual-license that older descriptions cite"** ? This is the subtle part. The first search result still described the **CODE** as GPL (an older/stale description), while the current README confirms code is now Apache-2.0. The claim correctly identifies that **the weights** are under OpenRAIL-M. The "GPL" in older descriptions referred to the CODE, not the weights ? and the weights have apparently been OpenRAIL-M throughout. The claim's framing ("NOT the GPL/commercial dual-license that older descriptions cite") is slightly loose because the GPL was the code license, not the weights license. But the claim's core thrust ? current weights = OpenRAIL-M, and code is now permissive Apache-2.0 ? is accurate.

The one place where the claim could be considered an *understatement* rather than overreach: OpenRAIL-M is not merely a "RAIL" use-restriction license here ? this *modified* version adds a hard **\$5M revenue/funding commercial cap**, which is a stronger restriction than vanilla OpenRAIL-M. This actually *strengthens* the underlying research point (weights are non-permissive/commercially restricted) rather than refuting the claim.

The claim is well-supported by the primary source, current as of the live README/HF card, and the source quality (the actual license files) matches the claim's strength. The minor looseness

about "GPL" referring to code-not-weights does not refute the central, well-evidenced assertion that code=Apache-2.0 and weights=OpenRAIL-M (restricted), and that checking only the repo LICENSE would mislead.

13. user (2026-05-31T04:25:25.500Z)

Structured output provided successfully

14. assistant (2026-05-31T04:25:31.828Z)

I've completed the adversarial verification.

Verdict: ****NOT REFUTED**** (refuted=false, high confidence).

The claim is well-supported by primary sources I verified directly:

- The repo `LICENSE` is genuinely Apache-2.0.
- The `MODEL\_LICENSE` is "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" and explicitly defines "Complementary Material" (source code/scripts) and excludes it from the license ("Complementary Material is not licensed under this License") ? matching the supporting quote.
- The current README and HF model card (`datalab-to/surya-ocr-2`, license tag `openrail`) both confirm code=Apache-2.0, weights=modified OpenRAIL-M.

The one looseness ? older sources citing "GPL" actually referred to Surya's **prior code** license (since changed to Apache-2.0), not the weights ? is immaterial and does not refute the central, primary-source-verified point. If anything, the modified OpenRAIL-M is **more** restrictive than the claim states (it adds a hard \$5M revenue/funding commercial cap), which reinforces rather than undermines the research conclusion that Surya's weights must be flagged as commercially unsafe by default.